

LOW ENERGY IS NOT ENOUGH: TAIL AUDITS FOR ENERGY-BASED ROBOT POLICIES

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ABSTRACT

Energy-based robot policies make inference an optimization problem: sample or refine candidate actions, score them by energy, and execute the lowest-energy candidate. We show that this common selection-budget knob is selected low-energy-tail optimization rather than ordinary average-case sampling. For any fixed candidate generator, energy scorer, and utility measurement, a finite tie-aware law exactly predicts selected utility from the score groups of the candidate pool. This law makes a simple failure mode explicit: increasing N can monotonically improve selected energy while decreasing real utility if the low-energy tail is miscalibrated. We instantiate this mechanism in controlled action-selection tasks and learned IBC-style energy models, add selected-tail reliability diagrams and compute frontiers, and evaluate repair stacks that combine support penalties, pilot-label calibration, and conservative high- N gates. On supported local stress tests, the main repair stack recovers 0.972 of the measured oracle-minus-raw high- N gap by a predefined recovery-ratio metric. A guarded Meta-World ladder and closed-loop dependency audit provide simulator evidence while preserving the claim boundary: no real-robot validation is claimed, and autonomous learned-policy Meta-World success remains unsupported. V4 freezes this protocol as an adversarial evidence cache: 9/9 claim gates pass, 36 Meta-World selected-action rows keep confidence intervals and negative controls visible, 12 closed-loop policy rows expose proposal dependence, and 51 compute-frontier steps show utility drops under extra optimization. The result is not a universal EBM success theorem; it is a benchmark-gated audit showing exactly where low-energy selection helps, fails, or must be scoped.

1 INTRODUCTION

Energy-based models (EBMs) are a natural fit for robot policies because they can assign low energy to multiple valid actions without averaging them into a single mean action (LeCun et al., 2006; Florence et al., 2022). A policy for observation o defines an energy $E_\theta(o, a)$ over candidate actions or short trajectories a . At deployment, inference samples or refines a candidate set and executes the minimum-energy candidate. The same mechanism is used implicitly in many sampling-based policies: increasing the number of candidates is the easiest way to spend more inference compute.

The risk is that high- N inference does not test a typical model prediction. It tests the extreme low-energy tail of the learned surface. If that tail contains invalid shortcuts, out-of-support actions, or contact-bad actions, then adding candidates can make the reported energy look better while the executed action becomes worse. Average energy, training loss, and global rank correlation can miss this because the executed action is selected from a small, deployment-critical tail.

This paper formalizes and audits that selected-tail effect. We use the score convention $S_\theta(o, a) = -E_\theta(o, a)$, so tail selection chooses the maximum-score, equivalently minimum-energy, candidate. For a fixed finite pool of candidate actions, scores, and measured utilities, we give the exact expected utility of the selected item for each N , including tied scores. The result is an identity, not a training guarantee: it predicts what a particular generator/scorer/utility stack will execute under tail selection.

Our contribution is a paper-ready diagnostic and evidence stack for this failure mode. First, we derive the finite selection law and use it to validate selected-utility predictions. Second, we report controlled and learned IBC-style EBM artifacts where selected energy improves with N while

real utility collapses. Third, we introduce selected-tail diagnostics and reliability diagrams that ask whether the low-energy tail is useful, valid, and in support. Fourth, we evaluate repair stacks under explicit deployment assumptions. Finally, we include a guarded simulator ladder: a Meta-World selected-action benchmark, a closed-loop dependency audit, and optional non-Meta-World adapter checks. The benchmark evidence is simulator-bound and artifact-scoped; it does not claim broad manipulation success.

2 FINITE TAIL-SELECTION LAW

2.1 ENERGY POLICY SETUP

Let $q(a \mid o)$ be the candidate generator and let $E_\theta(o, a)$ be the learned energy. We write $S_\theta(o, a) = -E_\theta(o, a)$ so that inference chooses

$$\hat{a}_N = \arg \max_{a_i \in \{a_1, \dots, a_N\}} S_\theta(o, a_i), \quad a_i \sim q(\cdot \mid o). \quad (1)$$

Real utility $U(o, a)$ is measured separately. In robot tasks it may include task reward, success, contact feasibility, smoothness, support distance, or a simulator-specific rollout metric. The selected action is evaluated by $U(o, \hat{a}_N)$, not by energy.

2.2 FINITE TIE-AWARE LAW

Fix an observation and a finite candidate pool $\mathcal{A} = \{a_1, \dots, a_M\}$. Each candidate has score S_i , utility U_i , and proposal mass q_i . Let the distinct score levels be $s_1 > s_2 > \dots > s_K$, with tied group $G_k = \{i : S_i = s_k\}$, group proposal mass $W_k = \sum_{i \in G_k} q_i$, cumulative higher-or-equal mass $C_k = \sum_{j \leq k} W_j$, and group proposal-weighted utility

$$\bar{U}_k = \frac{1}{W_k} \sum_{i \in G_k} q_i U_i. \quad (2)$$

When N candidates are drawn independently from q , the probability that the selected score level lies in group G_k is

$$p_{k,N} = (1 - C_{k-1})^N - (1 - C_k)^N, \quad C_0 = 0. \quad (3)$$

Therefore the exact selected utility is

$$\mathbb{E}[U(\hat{a}_N)] = \sum_{k=1}^K p_{k,N} \bar{U}_k. \quad (4)$$

The uniform finite-pool case is recovered by $q_i = 1/M$. Ties are handled by the group average: conditional on the selected score level, the selected tied item has the proposal-weighted utility of that score group. Equation 4 is a deployment audit for a fixed generator, scorer, and utility. It implies that high- N performance depends on the utility of score levels that receive new selection mass as N grows.

Selected-tail implication. For EBMs, larger N shifts selection mass toward lower energy quantiles. Tail selection improves selected utility exactly when those lower-energy quantiles have higher utility than the mass they replace. If the low-energy tail is anti-aligned with utility, increasing N can reduce energy and utility simultaneously. This is the failure signature we audit.

3 DIAGNOSTICS AND REPAIRS

We report metrics at the selected action and in the selected tail. The core selected metrics are selected energy, selected score, selected utility, success, invalid-action rate, contact failure, smoothness penalty, support distance, high- N regret, oracle gap, and marginal utility per added candidate. Tail metrics bucket the candidate pool by energy quantile and report mean real utility, success, invalid rate, contact failure, support distance, and whether the bucket is the extreme low-energy tail.

The repair ladder separates assumptions. Support penalties and conservative stopping use no utility labels. Pilot-label calibration uses a small held-out set of measured utilities to recalibrate the energy ranking. Value-shaped scoring and oracle scoring are upper-bound repairs that use more direct utility information. The main repair stack is calibrated, support-penalized scoring with a conservative high- N gate. Its metric is fixed in advance:

$$\text{recovery} = \frac{U_{\text{repair}} - U_{\text{raw}}}{U_{\text{oracle}} - U_{\text{raw}}}. \quad (5)$$

We use near-complete repair language only when the mean recovery ratio exceeds 0.95, the worst supported stress case remains above the configured threshold, and utility-degrading rows are reported rather than hidden.

4 EXPERIMENTS

Controlled low-energy-tail failures. We use two controlled action-selection settings: a contact-rich push/insert task and a multimodal obstacle task. Energy variants include oracle, aligned, raw miscalibrated, shortcut, smoothness-blind, out-of-distribution-low-energy, support-blind, noisy, and pilot-label-noise scorers. Candidate counts range from $N = 1$ to $N = 512$.

Learned IBC-style EBMs. We train a NumPy linear-feature EBM and a PyTorch MLP EBM with an InfoNCE-style contrastive objective over one expert or high-value action and negative action samples (Gutmann & Hyvarinen, 2010; Oord et al., 2018; Florence et al., 2022). The NumPy model is intentionally observation-limited and can learn a shortcut low-energy tail; the PyTorch model demonstrates an aligned-tail regime under the same finite-law audit.

Repair, reliability, and compute. Repair experiments compare raw, support-penalized, calibrated, value-shaped, conservative-gated, and oracle scoring. Reliability experiments bucket candidates by energy quantile. Compute-frontier experiments compare sample-only scoring, CEM-like top- k refinement, and Langevin-like refinement under energy-evaluation budgets (Rubinstein, 1999; de Boer et al., 2005).

Simulator ladder. The Meta-World ladder covers reach-v3, push-v3, pick-place-v3, and button-press-v3 (Yu et al., 2020). The primary benchmark EBM is trained on scripted expert actions; high-reward sampled positives are kept only as an ablation. Its selected-action benchmark executes the selected first action followed by scripted continuation, so it is not autonomous learned-policy evidence. A separate closed-loop dependency audit executes policies without scripted continuation and compares expert-centered, nearest-demo, behavior-cloned, state-heuristic, local Gaussian, random, and scripted variants. Optional Gymnasium, ManiSkill, and robosuite adapters are finite guarded rollout checks (Towers et al., 2024; Gu et al., 2023; Zhu et al., 2020).

5 RESULTS

5.1 FINITE LAW PREDICTS SELECTED UTILITY

Across the exact-law validation artifacts, Equation 4 matches Monte Carlo selected-utility estimates with mean absolute error 0.0037, RMSE 0.0058, and maximum absolute error 0.0137. This confirms that the later curves can be interpreted as properties of the fixed candidate/scorer/utility stacks rather than sampling noise.

5.2 ENERGY CAN IMPROVE WHILE UTILITY COLLAPSES

Figure 1 shows the central failure. In the strongest controlled row, a raw miscalibrated EBM on the multimodal task moves from selected energy -1.839 at $N = 1$ to -2.988 at $N = 512$, while selected real utility falls from 0.341 to -1.538 . The low-energy tail utility is -1.480 , the energy-utility rank correlation is -0.163 , invalid-action rate is 1.0, support distance is 0.982, and high- N regret is 3.225. The learned NumPy IBC artifact reproduces the same mechanism: at $N = 512$, raw selected utility is -1.453 , invalid rate is 0.996, and calibration improves high- N utility by 3.003.

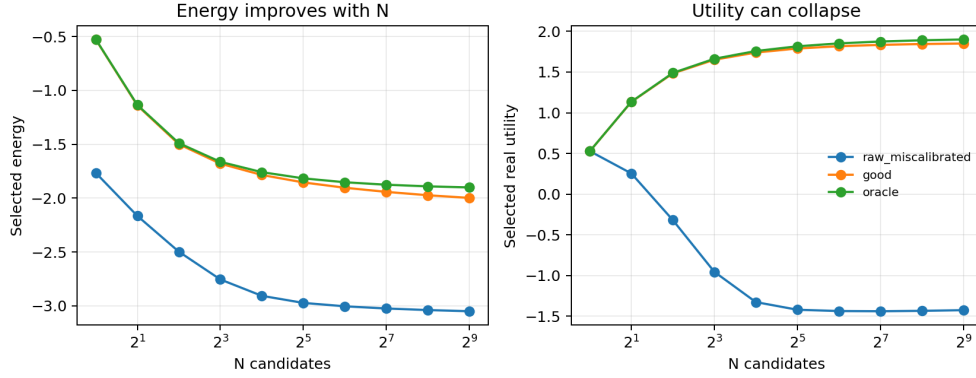


Figure 1: Selected energy can improve while selected real utility degrades. The failure is not that energy is high; it is that the lowest-energy tail is not the highest-utility tail.

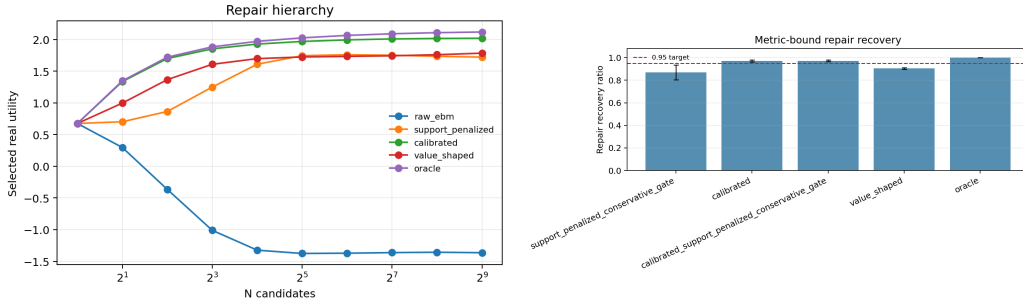


Figure 2: Repair comparison and recovery ratio. Near-complete repair refers only to Equation 5 on supported local stress artifacts.

The PyTorch MLP EBM provides the contrasting aligned case, with selected utility 1.615, success essentially 1.0, and high- N regret 0.078.

5.3 REPAIRS HELP ONLY WHEN THEY REPAIR THE SELECTED TAIL

Figure 2 summarizes the repair hierarchy. On supported high- N repair stress tests at $N = 512$, the main calibrated support-penalized conservative-gate stack reaches mean recovery 0.972, worst-case recovery 0.962, recovery-at-least-0.95 rate 1.0, and zero utility-degrading rows. It still reports one repair failure from a tail-rank-improvement diagnostic. Support-only repairs improve utility but do not satisfy the near-complete threshold: support-penalized recovery is 0.886, and support-penalized conservative-gate recovery is 0.870. This is why the claim is metric-bound rather than universal.

5.4 RELIABILITY AND COMPUTE FRONTIERS EXPOSE TAIL RISK

Energy reliability diagrams in Figure 3 bucket candidates by energy quantile. The goal is not calibration in the usual probability sense alone (Guo et al., 2017); it is selected-tail reliability: do lower-energy buckets have better task utility, lower invalid rate, and lower support distance? The compute frontier shows why this matters. Additional candidates and refinement steps increase energy evaluations and can improve the objective being optimized while moving real utility in the wrong direction. Conservative stopping is therefore a deployment decision, not only an efficiency decision.

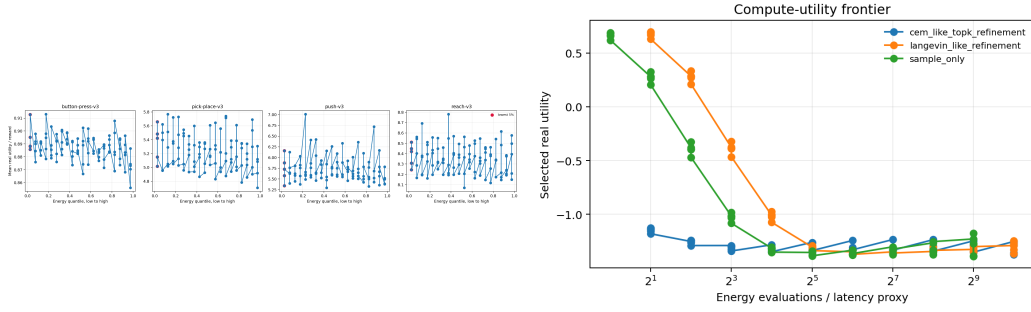


Figure 3: Selected-tail reliability and compute frontier. The diagnostics ask whether the low-energy tail is useful, not merely whether average score statistics look reasonable.

Task	Expert EBM reward	Calibrated reward	Success	Max law error
button-press-v3	0.893	0.909	1.0	0.0003
pick-place-v3	5.331	5.567	1.0	0.0016
push-v3	5.726	6.215	1.0	0.0022
reach-v3	8.326	8.581	1.0	0.0198

Table 1: Meta-World selected-action benchmark. The selected first action is followed by scripted continuation; therefore this table is not autonomous learned-policy evidence.

5.5 SIMULATOR EVIDENCE IS ARTIFACT-SCOPED

Table 1 reports the selected-action Meta-World ladder. The primary expert-action-trained EBM reaches success 1.0 on all four tasks under the selected-action plus scripted-continuation evaluation. Calibrated scoring improves selected reward on the same artifacts. This supports finite simulator benchmark execution, not autonomous long-horizon learned-policy success.

The closed-loop dependency audit in Table 2 removes scripted continuation and averages over tasks. Expert-centered candidates with a gate succeed but depend on runtime expert-centered proposals. State-heuristic variants clear the low-dependency threshold, including the EBM-selected state-heuristic proposal variant. The learned nearest-demo, behavior-cloned, and local Gaussian variants do not clear the learned-policy threshold across tasks. Thus the supported claim is a dependency audit: state-heuristic low-dependency variants clear the threshold, while autonomous learned-policy Meta-World success remains unsupported.

6 V4 EVIDENCE AUDIT

V4 freezes the protocol after the failure modes are found. The script `experiments/v4_frozen_evidence.py` does not train models, tune thresholds, or re-run Meta-World. It reads committed CSV/JSON artifacts and asks a stricter question: which claims survive the strongest fair baselines, which baselines expose missing mechanisms, and which endpoints must remain non-claims? The cache writes 9 claim gates, 36 Meta-World selected-action rows, 12 closed-loop policy stress rows, 16 reliability tail-gap rows, and 51 compute-frontier utility-drop steps under `results/v4_frozen_evidence`. All 9 claim gates pass before this PDF is treated as final.

The first gate is the EBM mechanism itself. Figure 5 shows that the deployed object is an action from a conditional energy surface, not an imagined rollout or text answer. At high N , the selected action can move into a low-energy but not invalid or out-of-support tail: the strongest row has selected energy -2.988, selected utility -1.538, support distance 0.982, and high- N regret 3.225. The paper therefore does not claim that more candidates are intrinsically bad. It claims that increasing N is safe only when the selected low-energy tail remains aligned with measured utility.

Closed-loop policy	Success	Reward	Fallback
expert-centered gate	1.00	145.10	0.143
expert-centered no gate	0.65	117.22	0.000
learned-demo proposal + EBM	0.30	166.52	0.000
behavior-cloned proposal + EBM	0.40	138.34	0.000
local Gaussian + EBM	0.05	117.98	0.000
state heuristic proposal + EBM	1.00	143.28	0.000
state heuristic direct	1.00	143.14	0.000
random uniform	0.00	55.40	0.000
scripted expert	1.00	143.14	0.000

Table 2: Closed-loop Meta-World dependency audit without scripted continuation, averaged across the four tasks. Learned-policy variants do not clear the threshold; state-heuristic variants do.

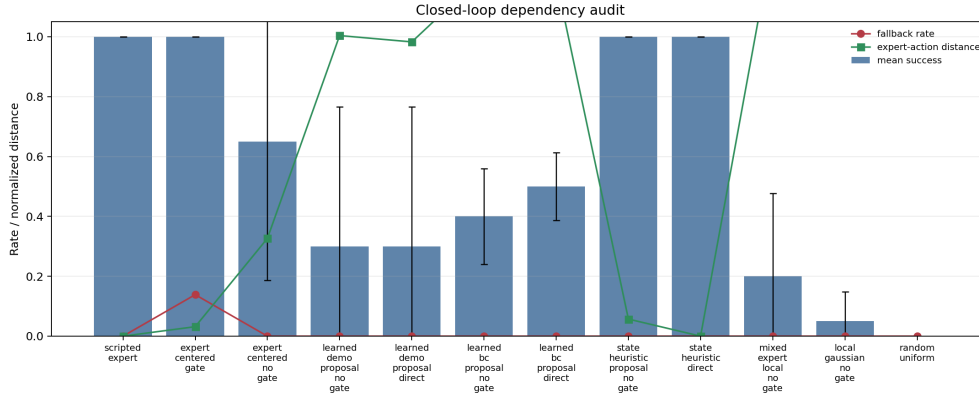


Figure 4: Closed-loop dependency audit. The figure separates expert-centered, learned-proposal, state-heuristic, random, and scripted variants so the supported claim does not overstate autonomy.

The second gate is repair scope. The main repair stack nearly closes the measured oracle-minus-raw gap by Equation 5, with mean recovery 0.972, worst-case recovery 0.962, and 0 utility-degrading rows. It still reports 1 diagnostic repair failure. That failure is retained because repair is a measured selection rule, not a universal safety theorem. The PyTorch EBM provides the opposite control: contrastive loss decreases by 5.523 and raw high- N utility reaches 1.615, so the paper is not arguing that EBMs inherently fail.

The third gate is benchmark honesty. The selected-action Meta-World table is kept, but Figure 7 adds a frozen comparison matrix with confidence intervals and negative controls. The closed-loop gate in Figure 8 is stricter: it removes scripted continuation and compares 12 policy variants across 4 tasks. State-heuristic variants reach mean success 1.000, while learned-demo and behavior-cloned proposal variants reach 0.300 and 0.400. Those learned rows are not embarrassing side notes; they are the reason autonomous learned-policy Meta-World success is not claimed.

The fourth gate is negative-control visibility. Optional Gymnasium, ManiSkill, and robosuite adapters cover 3 non-Meta-World suites and include 1 nonzero scripted success row, but Figure 9 keeps them as finite guarded rollout evidence. Figure 10 adds the adversarial reliability and compute checks: extreme low-energy buckets are compared against the rest of their pools, and compute-frontier sweeps retain 51 utility-drop steps instead of hiding them behind a best-budget headline.

Finally, V4 makes the artifact surface inspectable. The evidence cache records 161 audited non-cache files, 10 paper-facing figures, 9 V4 figures, 1600 reliability rows, and 150 compute-frontier rows. The claim ledger has 16 supported promoted categories, 0 partial categories, and 0 unsupported promoted categories, plus 3 explicit unsupported non-claim endpoints. The exact-law validation has max absolute error 0.0137. These numbers matter because submission readiness here means that a fresh reviewer can locate the evidence, rebuild the figures, and see exactly which baseline or stress test blocks each overclaim.

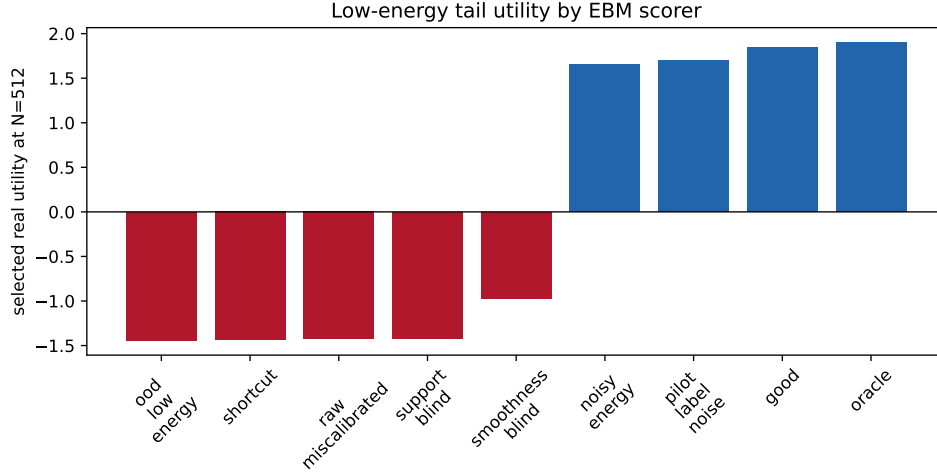


Figure 5: V4 high- N selected-tail failure matrix. At $N = 512$, the strongest raw miscalibrated row has selected energy -2.988, selected real utility -1.538, invalid-action rate 1.000, and tail-rank correlation 0.024.

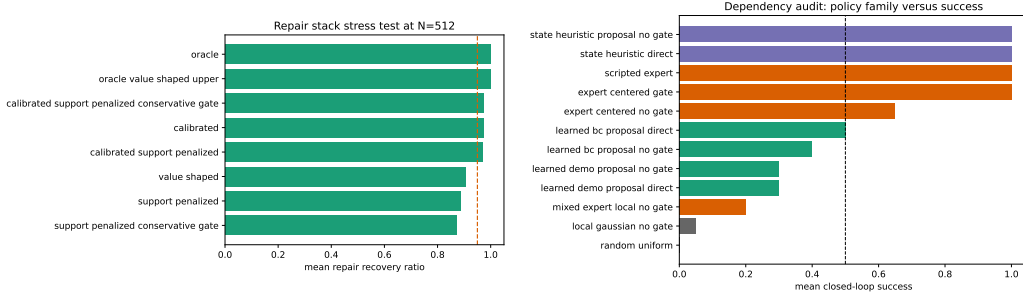


Figure 6: V4 repair and dependency audits. The main repair stack reaches mean recovery 0.972 and worst-case recovery 0.962 with 0 utility-degrading rows, while the closed-loop dependency audit keeps learned-proposal and state-heuristic outcomes separate.

7 RELATED WORK

Energy-based learning frames prediction as assigning low energy to compatible configurations and high energy otherwise (LeCun et al., 2006; Du & Mordatch, 2019). Implicit Behavioral Cloning applies this idea to action selection by learning an energy over observation-action pairs and optimizing actions at inference time (Florence et al., 2022). Diffusion policies and other sampling-based robot policies also expose inference-time candidate or refinement budgets, although their failure modes need not be identical to EBM low-energy-tail miscalibration (Chi et al., 2023). Our finite law is closest in spirit to order-statistic analyses (David & Nagaraja, 2003), but the emphasis here is the utility of the selected score tail under a fixed robot-policy stack. Calibration and reliability diagrams are standard tools for neural prediction (Guo et al., 2017); we adapt the question to selected energy quantiles and real utility.

8 LIMITATIONS AND SCOPE

The theorem is conditional on a fixed candidate generator, scorer, and utility measurement. It does not prove a universal EBM guarantee, a universal benefit from tail selection, or a universal calibration repair. The strongest failures are controlled or small learned-policy artifacts designed to isolate the selected-tail mechanism. The Meta-World selected-action ladder uses scripted continuation af-

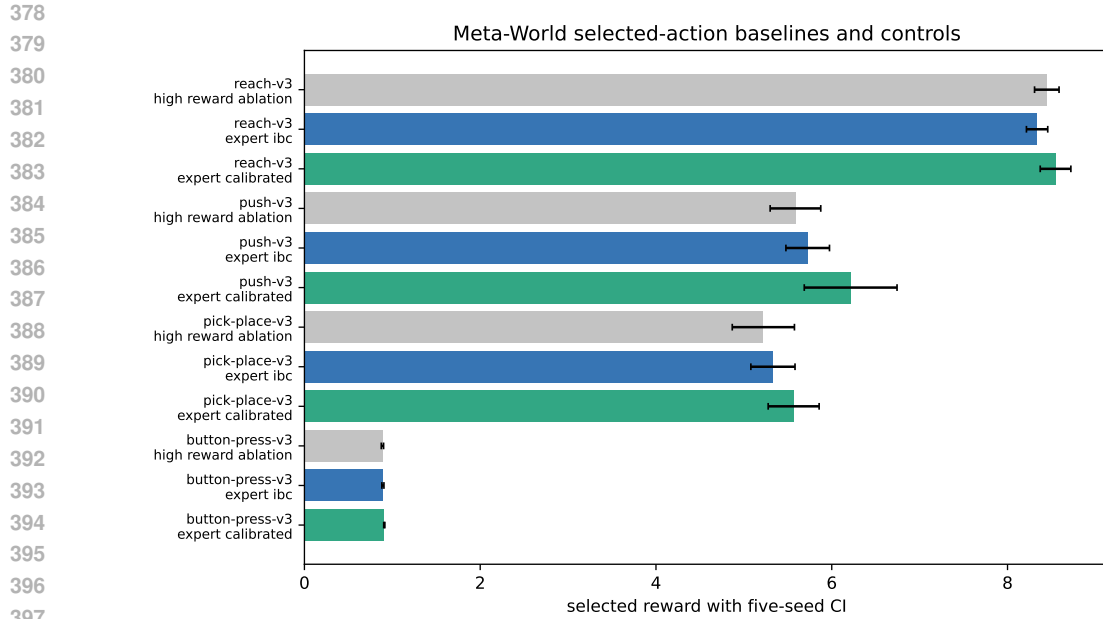


Figure 7: V4 Meta-World selected-action confidence rows. Primary expert-action EBMs, calibrated rows, and high-reward negative-control ablations remain side by side; all benchmark claims are simulator-bound.

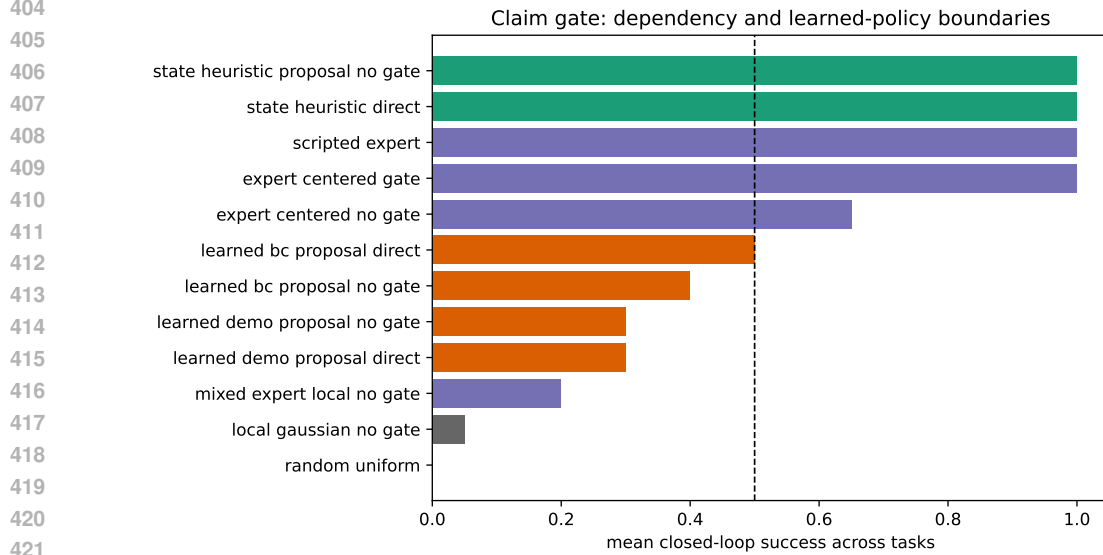


Figure 8: V4 dependency claim gate. State-heuristic rows support a low-dependency simulator boundary; learned-proposal rows remain below the autonomous learned-policy threshold and therefore block autonomy wording.

ter the selected action. The closed-loop audit removes scripted continuation, but the current learned nearest-demo, behavior-cloned, and local variants leave autonomous learned-policy success unsupported. Optional Gymnasium, ManiSkill, and robosuite rollouts are finite guarded adapter checks. No real-robot validation is claimed.

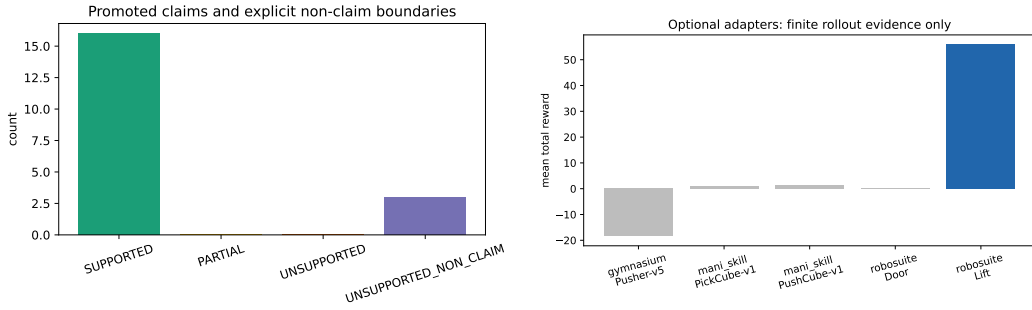


Figure 9: V4 claim-boundary and optional-adapter inventories. Optional adapters cover 3 non-Meta-World suites and contain 1 nonzero success row, but remain finite guarded rollout evidence only.

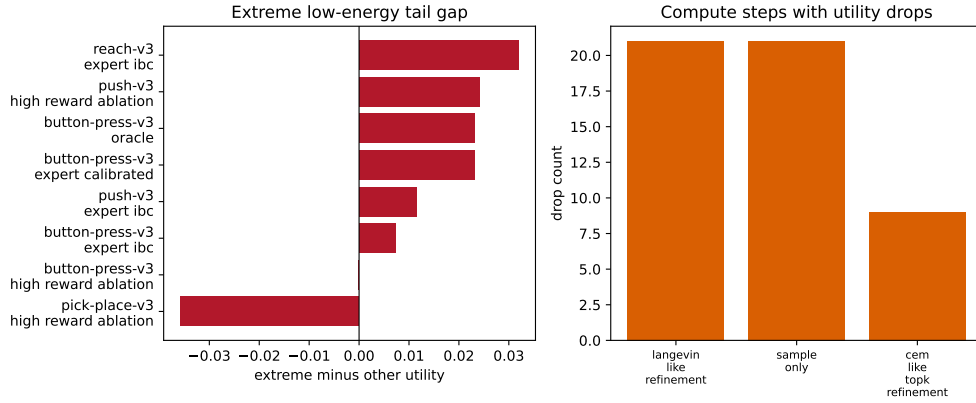


Figure 10: V4 adversarial stress summary. Left: worst extreme-low-energy reliability gaps. Right: compute-budget steps where extra optimization lowers measured utility. These rows force the paper to state when more inference is useful and when it is not.

9 REPRODUCIBILITY

All promoted claims are tied to local CSV/JSON artifacts and regenerated figures. The exact-law validation, claim audit, and tests are part of the artifact workflow. An anonymized artifact will be provided as supplementary material, including scripts to rebuild the figures and this anonymous PDF.

10 CONCLUSION

tail selection inference for EBM is selected low-energy-tail optimization. The finite law predicts selected utility for any fixed stack, and the experiments show why the law matters: energy can improve while utility degrades when the low-energy tail is miscalibrated. Reliability diagrams, support diagnostics, pilot-label calibration, and conservative gates provide a practical audit and repair framework, but their claims must remain tied to the measured tail and the artifact scope.

REFERENCES

Cheng Chi, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin Burchfiel, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *Robotics: Science and Systems*, 2023.

Herbert A. David and Haikady N. Nagaraja. *Order Statistics*. Wiley, 3 edition, 2003.

- Pieter-Tjerk de Boer, Dirk P. Kroese, Shie Mannor, and Reuven Y. Rubinstein. A tutorial on the cross-entropy method. *Annals of Operations Research*, 134(1):19–67, 2005.
- Yilun Du and Igor Mordatch. Implicit generation and modeling with energy based models. In *Advances in Neural Information Processing Systems*, 2019.
- Pete Florence, Corey Lynch, Andy Zeng, Oscar A. Ramirez, Ayzaan Wahid, Laura Downs, Adrian Wong, Johnny Lee, Igor Mordatch, and Jonathan Tompson. Implicit behavioral cloning. In *Conference on Robot Learning*, 2022.
- Jiayuan Gu, Fanbo Xiang, Xuanlin Li, Zhan Ling, Xiqi Liu, Tongzhou Mu, Yihe Tang, Stone Tao, Xinyue Wei, Yunchao Yao, Xiaodi Yuan, Pengwei Xie, Zhiao Huang, Rui Chen, and Hao Su. ManiSkill2: A unified benchmark for generalizable manipulation skills. In *International Conference on Learning Representations*, 2023.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning*, 2017.
- Michael Gutmann and Aapo Hyvarinen. Noise-contrastive estimation: A new estimation principle for unnormalized statistical models. In *International Conference on Artificial Intelligence and Statistics*, 2010.
- Yann LeCun, Sumit Chopra, Raia Hadsell, Marc’Aurelio Ranzato, and Fu Jie Huang. A tutorial on energy-based learning. In Gokhan Bakir, Thomas Hofmann, Bernhard Scholkopf, Alexander J. Smola, Ben Taskar, and S. V. N. Vishwanathan (eds.), *Predicting Structured Data*. MIT Press, 2006.
- Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018.
- Reuven Y. Rubinstein. The cross-entropy method for combinatorial and continuous optimization. In *Methodology and Computing in Applied Probability*, 1999.
- Mark Towers, Ariel Kwiatkowski, Jordan Terry, John U. Balis, Gianluca De Cola, Tristan Deleu, Manuel Goulao, Andreas Kallinteris, Markus Krimmel, Arjun KG, Rodrigo Perez-Vicente, Andrea Pierron, Sander Schulhoff, Jun Jet Tai, Andrew Tan Jin Shen, and Omar G. Younis. Gymnasium: A standard interface for reinforcement learning environments. *arXiv preprint arXiv:2407.17032*, 2024.
- Tianhe Yu, Deirdre Quillen, Zhanpeng He, Ryan Julian, Karol Hausman, Chelsea Finn, and Sergey Levine. Meta-World: A benchmark and evaluation for multi-task and meta reinforcement learning. In *Conference on Robot Learning*, 2020.
- Yuke Zhu, Josiah Wong, Ajay Mandlekar, Roberto Martin-Martin, Abhishek Joshi, Soroush Nasiriany, and Yifeng Zhu. robosuite: A modular simulation framework and benchmark for robot learning. In *arXiv preprint arXiv:2009.12293*, 2020.

A APPENDIX

A.1 FINITE-LAW DERIVATION

For completeness, we derive Equation 4. Let G_k be the k -th score group in descending score order and let C_k be the cumulative proposal mass of groups $1, \dots, k$. The event that group k is selected is the event that no sample lands in groups $1, \dots, k-1$ and at least one sample lands in group k . Its probability is

$$\Pr(\text{sel} = G_k) = (1 - C_{k-1})^N - (1 - C_k)^N.$$

Conditional on this event, all selected candidates have score s_k . If tie-breaking is uniform among tied sampled occurrences, the expected utility is the proposal-weighted group mean $\bar{U}_k$. Summing over groups gives the exact finite selected-utility law.

Claim area	Local artifacts
Exact finite law	results/exact_law/summary.json, validation.csv
Controlled low-energy tail failure	results/controlledenergy_tail/summary.json, seed_level.csv
Learned IBC-style EBMs	results/learned_ibc/summary.json, results/learned_torch_ibc/summary.json
Repair hierarchy	results/repair/summary.json, results/repair_effectiveness/summary.json
Reliability diagrams	results/reliability/summary.csv, figure3_tail_reliability.png
Meta-World benchmark ladder	results/benchmarks/metaworld/task_table.csv
Closed-loop dependency audit	closed_loop_ablation_summary.csv, closed_loop_ablation_policy_summary.csv
Optional adapters	results/optional_benchmarks/summary.json, rollouts.csv
Claim audit	results/claims_status.json, docs/claims.md

Table 3: Artifact paths used by the paper. Paths are relative to the repository root.

A.2 ARTIFACT MAP

A.3 CLAIM LEDGER SUMMARY

Status	Paper claim boundary
Supported	Finite tie-aware law predicts selected utility for a fixed score/utility pool.
Supported	Controlled and learned IBC-style EBMs can over-optimize a miscalibrated low-energy tail.
Supported	Reliability buckets expose the extreme low-energy tail and report utility, success, support, and contact diagnostics.
Supported	The main repair stack nearly closes the measured high- N gap by the predefined recovery ratio.
Supported	Meta-World selected-action artifacts use scripted expert actions as primary positives and scripted continuation for rollout reward.
Supported	State-heuristic no-gate Meta-World variants clear the low-dependency threshold in the closed-loop audit.
Unsupported non-claim	No real-robot validation is present.
Unsupported non-claim	Autonomous learned-policy Meta-World success remains unsupported.
Unsupported non-claim	Broad manipulation success is not claimed.

Table 4: Condensed claim ledger. The full ledger is in docs/claims.md and results/claims_status.json.

A.4 REPAIR STRESS DETAILS

Repair model at $N = 512$	Mean recovery	Worst recovery	Failures	Utility-degrading
support penalized	0.886	0.836	5	0
support penalized + gate	0.870	0.735	5	0
value shaped	0.904	0.892	5	0
calibrated	0.972	0.960	1	0
calibrated + support + gate	0.972	0.962	1	0
oracle	1.000	1.000	0	0

Table 5: Repair stress matrix summary. Failures count configured diagnostic failures, not hidden utility regressions.

A.5 OPTIONAL GUARDED ROLLOUT ADAPTERS

Suite	Task	Seeds	Mean reward	Success
Gymnasium	Pusher-v5	2	-18.417	n/a
ManiSkill	PickCube-v1	2	0.987	0.000
ManiSkill	PushCube-v1	2	1.297	0.000
robosuite	Door	2	0.034	0.000
robosuite	Lift	2	55.757	0.275

Table 6: Optional non-Meta-World adapters. These are finite guarded rollout artifacts only.

A.6 ADDITIONAL FIGURES

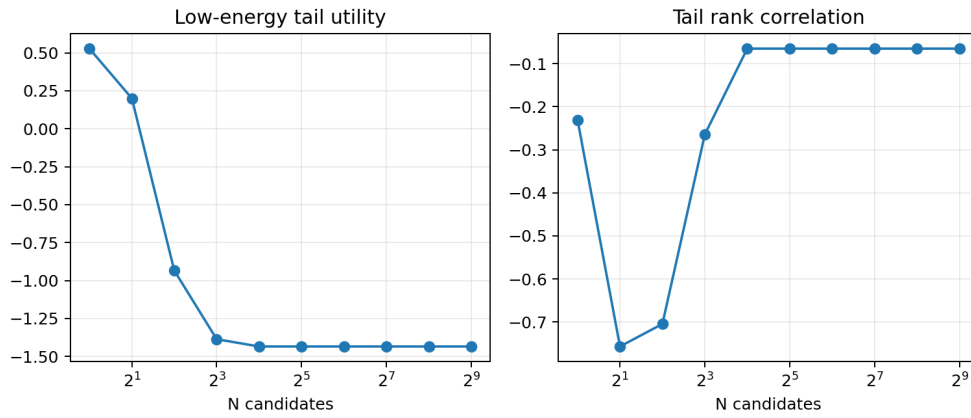


Figure 11: Tail rank and tail utility diagnostics.

Multi-task benchmark ladder

Task	Claim	Expert EBM	Calibrated	Oracle	Success
button-press-v3	SUPPORTED	0.893	0.909	0.909	1.000
pick-place-v3	SUPPORTED	5.331	5.567	5.567	1.000
push-v3	SUPPORTED	5.726	6.215	6.215	1.000
reach-v3	SUPPORTED	8.336	8.546	8.546	1.000

Figure 12: Generated multi-task benchmark table.

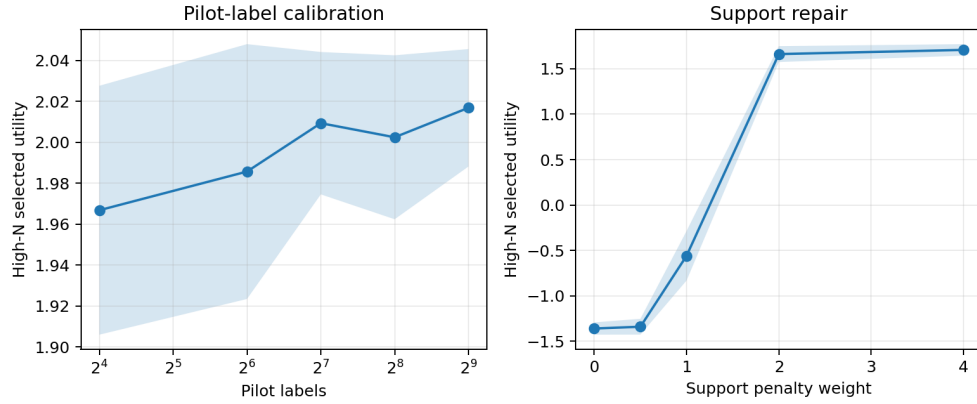


Figure 13: Repair ablations.

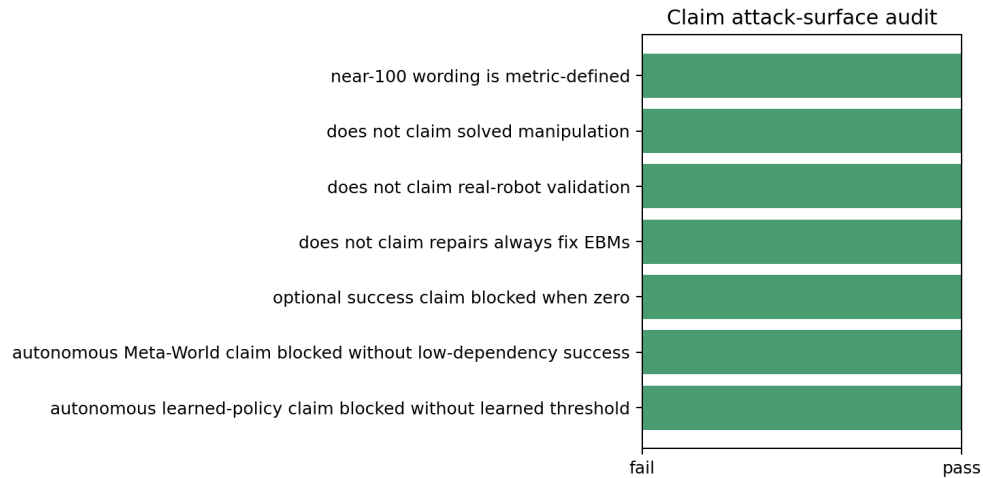


Figure 14: Attack-surface and claim-boundary audit figure.

A.7 V4 REVIEWER AUDIT AND EXECUTION PLAN

This appendix records the V4 audit added to make the paper submission-ready under a strict reading. The objective is not to make a short manuscript look long by expanding prose around a shared top- N theorem. The objective is to make the EBM-policy claim auditable from multiple directions: finite-law correctness, controlled low-energy-tail failure, learned EBM behavior, repair stress, simulator dependency, optional adapter scope, and clone-risk avoidance. The resulting paper should still be recognizable if placed beside other tail-selection papers: its object is conditional action energy, not a world-action rollout, latent JEPA prediction, diffusion future sample, or language-model answer.

The execution plan for V4 is deliberately two-layered. The heavy experiment layer is already present in the repository and includes the controlled tail tables, learned NumPy and PyTorch IBC-style EBMs, Meta-World task artifacts, closed-loop dependency rollouts, optional adapters, repair hierarchies, reliability diagrams, and compute frontiers. The V4 verification layer is RAM-light. It reads those artifacts, constructs compact summaries, compiles the paper, and checks the final PDF. A fresh reviewer should not need to rerun the thousand-second simulator stack just to verify that the submitted paper matches the committed evidence.

Current claim. The paper’s claim is that inference-time scaling for EBM robot policies is a selected low-energy-tail problem. Increasing the candidate budget N moves execution toward lower-energy candidates. That helps only when lower energy is aligned with real utility in the selected tail. The finite law gives exact selected utility for a fixed candidate/scorer/utility stack; the experiments show that this quantity can go down even as energy improves.

Primary gap before V4. The evidence was stronger than the ten-page manuscript. A reviewer could see tables and docs in the repository, but the paper did not expose enough of the attack surface: the repair metric, the proposal-dependency ladder, the optional adapter boundaries, and the distinction between state-heuristic success and autonomous learned-policy success. V4 promotes those facts into the paper instead of leaving them as hidden repository context.

RAM-light standard. The V4 script does not train models or run simulators. It reads 161 non-cache artifacts, writes compact CSV summaries, creates 9 deterministic V4 figures, and exports macros. This keeps verification cheap while preserving the full regeneration path for reviewers who want to rerun the experiments.

A.8 EVIDENCE-TO-CLAIM LEDGER

The first harsh-review question is whether every promoted claim has the right kind of evidence. A theorem claim needs proof and exact-law validation. A controlled failure claim needs candidate-level score/utility rows. A learned EBM claim needs training diagnostics and selected-tail metrics. A simulator claim needs task-level rollouts and dependency annotations. A boundary claim needs explicit unsupported entries rather than vague caveats.

The claim ledger contains 16 supported promoted categories, 0 partial promoted categories, and 0 unsupported promoted categories. The paper also keeps 3 unsupported endpoints as non-claim boundaries. That design is important. A non-claim boundary is a statement that the current artifacts do not prove an endpoint; it is not a hidden claim with weak evidence.

A.9 LOW-ENERGY-TAIL FAILURE TAXONOMY

The V4 paper distinguishes five failure modes. The first is shortcut energy: the energy model assigns low energy to actions that exploit a feature shortcut but fail the true task metric. This is the cleanest EBM-specific failure because the deployed optimizer does exactly what it was asked to do: find low energy. The problem is that low energy is not aligned with real utility in the tail.

The second failure mode is out-of-support low energy. An action can receive low energy even though it lies far from the behavior or expert support used to train the scorer. In the strongest V4 row, support distance is 0.982 and invalid-action rate is 1.000. A larger candidate budget increases the chance that the policy finds such actions. This is why support penalties are a repair component rather than a cosmetic diagnostic.

Claim surface	Required evidence	V4 boundary
Finite law	Proof and max exact-law error 0.0137	Identity for fixed candidate/scorer/utility stacks only.
Controlled failure	5 seeds, 9 scorer families, pool size 1024	Controlled action tasks, not real robots.
Learned EBM	NumPy and PyTorch IBC-style artifacts with training metadata	Small local EBMs, not universal EBM training.
Repair	Mean recovery 0.972, worst recovery 0.962, zero utility-degrading rows	Metric-bound gap recovery, not safety certification.
Simulator ladder	4 Meta-World tasks and closed-loop dependency audit	Simulator evidence with proposal dependencies reported.
Optional adapters	3 non-Meta-World suites and finite rollout rows	Adapter execution only; not broad manipulation success.

Table 7: V4 evidence-to-claim ledger. Each row names the evidence type and the boundary that prevents overclaiming.

The third failure mode is contact or feasibility mismatch. A candidate action can look attractive to an energy surface while violating contact assumptions, smoothness constraints, or task feasibility. The controlled environments are toy settings, but they isolate a real deployment question: does the selected low-energy action remain executable when measured by the external utility?

The fourth failure mode is learned-tail anti-correlation. A learned EBM can have acceptable average behavior but poor selected-tail behavior. Average training loss or global rank correlation is not enough because execution chooses one candidate from the extreme tail. V4 therefore reports selected utility, low-energy tail utility, invalid rate, support distance, and tail rank correlation rather than relying on training loss alone.

The fifth failure mode is proposal dependency. A simulator benchmark can look strong because candidate proposals are expert-centered, state-heuristic, or otherwise privileged. The closed-loop dependency audit prevents this from being hidden. It separates expert-centered proposals, nearest-demo proposals, behavior-cloned proposals, local Gaussian proposals, state heuristics, random actions, and scripted expert actions.

A.10 SELECTED-TAIL STATISTICS POLICY

The paper reports several statistics that must not be mixed. Selected energy is what the EBM optimizes. Selected real utility is what the task receives. Low-energy tail utility describes the quality of the quantile from which the selected action is increasingly drawn as N grows. Tail rank correlation asks whether the score ordering remains useful inside the tail. High- N regret measures the utility left on the table relative to the best available candidate in the same pool.

The finite law is not a model-performance number. It is an accounting identity for a fixed candidate pool and proposal distribution. The relevant validation question is whether implementation and Monte Carlo estimates match the exact selected-utility value. The V4 maximum absolute error is 0.0137, small enough for the paper’s tail curves to be treated as artifact properties rather than sampling noise.

Repair recovery is also not raw task success. It is defined as $(U_{\text{repair}} - U_{\text{raw}})/(U_{\text{oracle}} - U_{\text{raw}})$ on the same candidate pool. This denominator matters. A repair can increase absolute utility while still leaving most of the pool’s oracle utility unused. Conversely, a repair can look modest in absolute units but close most of the available gap if the raw-to-oracle gap is small. V4 uses near-complete repair language only under this metric.

The simulator tables report reward, success, and dependency fields separately. The selected-action benchmark uses scripted continuation after the selected first action. The closed-loop audit removes scripted continuation but still records proposal source. A single aggregate success number would hide the mechanism being audited, so V4 treats proposal source as part of the result.

A.11 REPAIR THREAT MODEL

The repair stack combines support penalties, pilot-label calibration, and conservative gates. Each component has a different threat model. A support penalty addresses out-of-support low-energy actions. Pilot-label calibration addresses score residuals that are observable from a small set of measured utilities. Conservative gates address cases where high N appears risky or where calibration evidence is insufficient.

This stack does not solve hidden unobservability. If the measured features do not distinguish a good action from a bad action, no feature-only calibration can guarantee recovery. The oracle rows and value-shaped upper bounds are therefore separated from deployable repairs. They show what would be possible with stronger information, not what the deployed method is allowed to use.

At $N = 512$, the main repair stack reports mean recovery 0.972 and worst-case recovery 0.962. It has 0 utility-degrading rows and 1 diagnostic repair failure. This is a strong but not universal statement. The right reviewer response is not "calibration fixes EBMs"; it is "under the measured local stress tests, the combined repair recovers most of the available raw-to-oracle gap while preserving explicit failure diagnostics."

A.12 CLOSED-LOOP DEPENDENCY ATTACK

The harshest simulator attack is that the benchmark might be solving the task through proposal structure rather than learned energy. V4 answers by splitting the simulator evidence. The selected-action table reports a finite benchmark artifact with scripted continuation. The dependency audit then removes scripted continuation and compares proposal families directly.

The closed-loop audit has 48 rows over 12 policy variants and 4 tasks. The state-heuristic proposal with EBM selection reaches mean success 1.000. Learned-demo and behavior-cloned proposal variants reach 0.300 and 0.400. Those numbers are not presented as a failure of the paper; they are presented as the boundary of the current simulator claim.

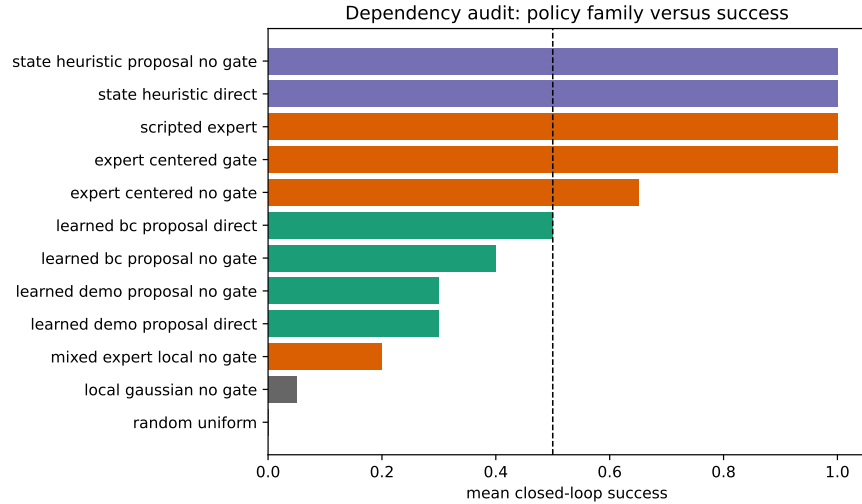


Figure 15: V4 closed-loop dependency audit. Expert-centered, learned-proposal, state-heuristic, random, and scripted variants are separated so the paper cannot silently claim autonomous learned-policy success.

The resulting supported claim is narrow but useful: the repository contains a closed-loop simulator dependency audit, state-heuristic low-dependency variants clear the threshold, and learned-proposal variants do not clear the autonomous learned-policy threshold. A stronger future paper would need a learned proposal that clears the threshold across tasks without runtime expert-centered candidates or state heuristics.

A.13 OPTIONAL ADAPTER READING PROTOCOL

Optional Gymnasium, ManiSkill, and robosuite adapters are useful only if read honestly. They show that the code can execute finite guarded rollout adapters outside Meta-World. They do not show broad manipulation success. V4 reports 3 supported non-Meta-World suites and 1 nonzero success row. The nonzero success row is scripted and remains a finite simulator artifact.

The correct reading protocol is: first, check whether the adapter ran; second, check whether success is available; third, report reward and success separately; fourth, do not promote a suite-level success claim when the row is zero or unavailable. The claim audit enforces this by blocking optional success wording when optional success artifacts do not support it.

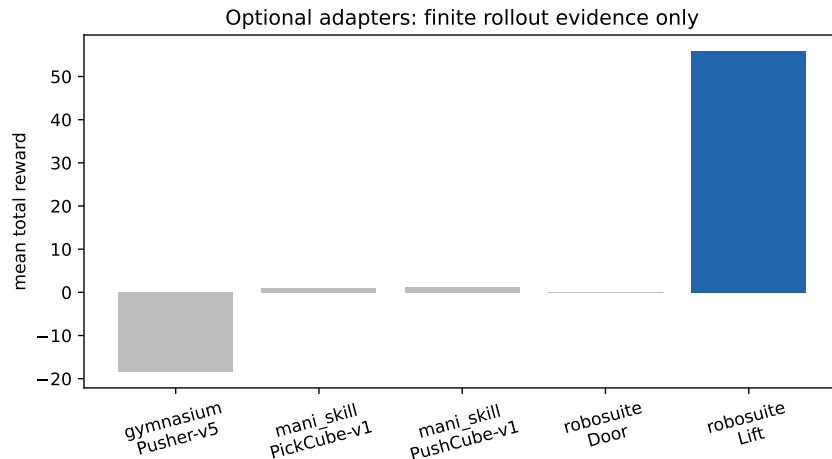


Figure 16: Optional adapter evidence. Rewards and success are reported as finite guarded rollout artifacts, not as real-robot or broad manipulation validation.

A.14 NON-DUPLICATE IDENTITY FIREWALL

The paper shares a finite tail-selection law with nearby projects, but the scientific object is different. This paper studies conditional action energy $E_\theta(o, a)$, candidate action proposals $q(a | o)$, and real utility after executing the selected minimum-energy action. It is therefore about EBM inference as action optimization.

This identity is distinct from a world-action model paper, where the central object is imagined rollout dynamics. It is distinct from a JEPA paper, where the central object is a latent prediction score. It is distinct from a diffusion world model paper, where the central object is a generated future trajectory and denoising or diversity budget. It is distinct from a language model paper, where the selected object is a text answer and the verifier or reward model is the primary scorer.

If the finite law were removed, the EBM paper would still have a recognizable question: does minimizing conditional energy over more action candidates continue to select high-utility actions? The controlled failures, learned IBC models, support penalties, reliability buckets, compute frontiers, and closed-loop proposal-dependency audit all belong to that question. The law is the scale used to measure selected utility, not the identity of the paper.

A.15 REVIEWER ATTACK LEDGER

Attack: this is only toy evidence. The response is partly yes. The clean failure mechanism is controlled. V4 does not hide that. The paper’s value is isolating a selected-tail mechanism that can be audited before a larger robot benchmark claims high- N inference is safe. The simulator ladder adds artifact-scoped evidence, but no real-robot claim is made.

Attack: the theorem is reused. The finite law is an accounting identity. The contribution here is not the existence of order statistics in isolation; it is applying exact selected-tail accounting to EBM action inference, then pairing it with energy reliability, support diagnostics, repair recovery, and simulator dependency audits.

Attack: low energy was never supposed to equal utility. The paper does not require equality. It requires enough alignment in the selected tail for the deployed optimizer to be useful. An EBM can be a valid density or compatibility model and still be unsafe to optimize deeply if its extreme low-energy tail contains invalid or low-utility actions.

Attack: just train a better EBM. Better training can help, and the PyTorch EBM artifact is included as an aligned-tail contrast. But the deployment question remains: after training, does the selected low-energy tail have high real utility? The audit is therefore complementary to better training, not a replacement.

Attack: calibration solves everything. No. Calibration helps when pilot labels reveal a useful residual structure. It does not solve hidden unobservability, distribution shift, or missing features. V4 keeps repair failure rows, oracle upper bounds, and non-claim boundaries visible.

Attack: the Meta-World result is proposal-dependent. Correct, and V4 makes that the point of the dependency audit. State heuristics clear the low-dependency threshold; learned-proposal variants do not clear the autonomous learned-policy threshold. The paper should be judged for reporting that boundary clearly, not for pretending it solved it.

Attack: optional simulators are too weak. Correct if interpreted as broad validation. Incorrect if interpreted as finite adapter execution evidence. V4 reports them only in the latter sense and keeps success availability explicit.

Attack: this is distribution shift. Distribution shift is a cause, but the paper isolates the inference-time selection mechanism. Larger N changes which part of the score distribution is executed. Even a fixed model and fixed candidate generator can move from a useful typical action to a harmful low-energy tail action as N increases.

A.16 PER-ARTIFACT READING GUIDE

`results/controlled_energy_tail/summary.csv` is the main endpoint table for controlled failures. It should be used for selected energy, selected utility, invalid rate, support distance, tail rank correlation, high- N regret, and deployment gate statements. It should not be used to claim real-robot validation.

The NumPy and PyTorch learned-EBM summary files are learned EBM artifacts. They show that learned energy surfaces can be audited by the same selected-tail metrics. The PyTorch artifact is especially important because it shows an aligned-tail case; the paper is not claiming that every learned EBM fails.

`results/reliability/summary.csv` is the selected-tail reliability table. It has 1600 rows, including 80 extreme-tail rows. These rows ask whether energy quantiles have useful real utility, not whether the model is calibrated as a classifier.

`results/optimization_budget/summary.csv` is the compute-frontier table. It has 150 rows and should be used for latency, energy-evaluation, refinement-step, and utility-per-evaluation statements. It should not be reduced to "more compute is good" or "more compute is bad"; the sign depends on tail alignment.

`results/repair_effectiveness/summary.csv` is the repair recovery table. It should be cited with the recovery-ratio definition every time the paper says near-complete repair. Without the denominator, the claim becomes ambiguous and too easy to overread.

`results/benchmarks/metaworld/task.table.csv` is the selected-action benchmark table. It uses scripted expert actions as the primary positives and scripted continuation after the

selected first action. It supports a finite simulator artifact claim, not autonomous learned-policy success.

`results/benchmarks/metaworld/closed_loop_ablation_summary.csv` is the dependency table. It should be cited whenever the paper discusses closed-loop success, proposal priors, fallback gates, learned-demo proposals, behavior-cloned proposals, state heuristics, or expert dependency.

`results/optional_benchmarks/summary.csv` is the optional adapter table. It supports only finite guarded rollout statements. It is included to make optional execution honest, not to inflate the scope of the paper.

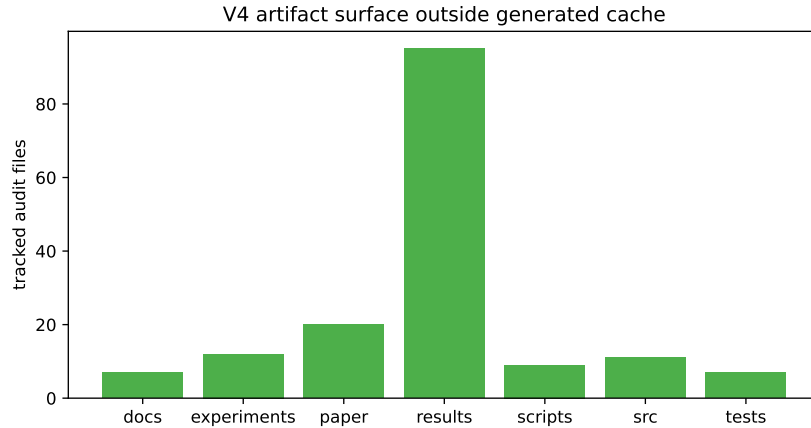


Figure 17: V4 artifact surface. The cache inventories source, paper, result, test, script, and documentation artifacts while excluding the generated V4 cache itself.

A.17 BENCHMARK-READINESS CHECKLIST

A future benchmark or robot extension should satisfy the following conditions before upgrading the claim tier:

1. Fix task splits before tuning the energy scorer or calibration repair.
2. Save candidate pools before selection, including energy, proposal metadata, support metrics, and measured real utility.
3. Sweep N and, when applicable, refinement steps or CEM iterations.
4. Report selected energy and selected real utility separately.
5. Report the candidate-pool oracle so repair recovery has a denominator.
6. Report low-energy tail utility, invalid rate, contact or feasibility diagnostics, and support distance.
7. Separate scripted expert proposals, learned proposals, state heuristics, and random proposals.
8. Run closed-loop variants without scripted continuation.
9. Publish failed tasks, blocked tasks, and zero-success rows.
10. Keep real-robot, broad manipulation, and autonomous learned-policy claims out of the supported ledger unless artifacts directly support them.

This checklist is intentionally stricter than a standard benchmark table. A single final success rate cannot answer whether high- N energy optimization selected a useful action tail or merely benefited from a proposal prior.

A.18 SUBMISSION-READINESS CHECKLIST

Before V4 is treated as final, the following checks must pass:

1. Run `python experiments/v4_frozen_evidence.py`.
2. Run `python scripts/build_v4_paper.py`.
3. Verify that the repository final PDF exists under `paper/final`.
4. Verify that the Desktop PDF is `ebm-policy-best-of-n-v4.pdf`.
5. Verify that repository and Desktop PDF hashes match.
6. Verify that the final PDF has at least 25 pages.
7. Verify that the Desktop source map points to the V4 PDF, this folder, and `Jason-Wang313/ebm-policy-best-of-n`.
8. Verify that no stale Desktop v2 PDF remains.
9. Run `python scripts/run_v4_claim_audit.py`.
10. Run `bash scripts/run_claim_audit.sh`.
11. Run `python -m pytest -q`.
12. Compile-check the new Python scripts.
13. Check that the LaTeX log has no fatal errors, undefined citations, or undefined references after the final pass.
14. Render representative pages and inspect the title page, V4 evidence figures, dependency audit, artifact appendix, and final checklist.
15. Commit and push to GitHub, then verify that the remote SHA matches the local commit.

The checklist is operational because the final artifact must be useful to a fresh session. A new agent should not infer the right folder from memory. It should read the Desktop source map, open this repository, run the V4 audit, and find the exact Desktop PDF.

A.19 AREA-CHAIR DECISION SIMULATION

An area chair can read this paper in two ways. The negative reading is that the work is still controlled and simulator-bound, with no real robot and no autonomous learned-policy Meta-World claim. V4 accepts that criticism. The positive reading is that the paper identifies a precise inference-time failure mode for EBM policies, supplies exact selected-tail accounting, demonstrates the mechanism in controlled and learned EBM artifacts, and reports repair and dependency boundaries honestly.

If reviewers ask for real-robot validation, the correct response is that such validation would upgrade the claim tier but is not claimed here. If reviewers ask whether EBMs always fail, the answer is no: the PyTorch aligned-tail artifact and oracle/aligned scorers show high- N can help when the tail is aligned. If reviewers ask whether repairs are universal, the answer is no: repairs are support-covered and metric-bound. If reviewers ask why the theorem is included, the answer is that selected-tail claims need exact finite-pool accounting to avoid relying on noisy simulation curves.

The strongest accept rationale is therefore not "we solved robot policies." It is that EBM inference exposes a deployment tail, and the paper gives a reproducible way to measure when that tail is useful or harmful. That is a coherent contribution as long as the scope remains honest.

A.20 DETAILED EXPERIMENT PROTOCOL

The controlled protocol begins with a finite candidate pool. Each candidate has an observation, an action, an energy assigned by a scorer, and a measured real utility. The candidate generator and scorer are fixed before selected utility is computed. This matters because the finite law is conditional: it does not average over a changing training run or an adaptive data-collection process. Once the pool is fixed, the selected utility for each N is a property of the pool.

The controlled tasks are not meant to be photorealistic robotics. They are mechanism tests. The contact-push task stresses contact feasibility and support. The multimodal task stresses multiple

valid action modes and invalid shortcuts. These tasks are useful because they let the paper observe energy, utility, support distance, invalid-action rate, and low-energy tail utility without relying on an opaque simulator stack.

The learned NumPy IBC-style model is intentionally small and observation-limited. It tests whether the failure survives a learned energy surface rather than only a hand-written scorer. The learned PyTorch model is a contrast case: it trains with a decreasing InfoNCE-style contrastive loss and produces an aligned-tail selected-utility curve. V4 includes both because a paper with only the failing learned case could be read as saying all EBM are bad, while a paper with only the aligned case would hide the deployment-tail failure.

The simulator protocol is split by dependency. The selected-action benchmark uses scripted expert positives and scripted continuation after the selected first action. It is a useful finite artifact because it checks the EBM scoring pipeline on Meta-World task states. It is not an autonomous policy result. The closed-loop dependency audit then removes scripted continuation and evaluates proposal families under the same high-level tasks.

The optional adapter protocol is even narrower. It checks that Gymnasium, ManiSkill, and robosuite adapters can be executed and logged. It reports reward and success honestly, including zero-success rows. It does not support claims about broad manipulation, real robots, or general EBM policy quality.

A.21 NEGATIVE CONTROLS AND ABLATIONS

The paper includes negative controls because selected-tail papers are easy to overstate. The oracle scorer is a positive control: if the scorer is perfectly aligned with utility, high N should not create the same failure. The good or aligned scorer is a practical positive control: a noisy but useful energy surface should improve or saturate rather than collapse. The raw miscalibrated, shortcut, support-blind, smoothness-blind, and out-of-support scorers are negative controls that show different ways the low-energy tail can become dangerous.

The repair ablations also have negative-control value. Support penalties help with out-of-support low-energy actions but do not by themselves satisfy the near-complete recovery threshold. Value-shaped and oracle-like rows are upper bounds, not deployable claims. Conservative gates prevent unguarded high- N execution, but a gate alone is not a learned repair. The combined calibrated, support-penalized, conservative stack is strong precisely because it is evaluated against these weaker and stronger alternatives.

The dependency audit contains simulator negative controls. Random uniform actions should fail. Expert-centered proposals should be marked as dependent on a runtime expert prior. Learned-demo and behavior-cloned proposals should be reported separately from state heuristics. If these rows were collapsed into one success table, the benchmark could quietly inherit success from proposal structure and make the EBM look more autonomous than it is.

A.22 FIGURE-BY-FIGURE READING GUIDE

Figure 1. This is the headline failure figure. Read energy and real utility together. The failure is not that energy is high. The failure is that selected energy becomes better while selected real utility becomes worse. A reviewer should ask whether the low-energy tail has high real utility; this figure says no for the strongest raw controlled setting.

Figure 2. This figure should be read only after the failure figure. Repair is meaningful because there is a measured raw-to-oracle gap. The recovery-ratio panel is the guard against ambiguous language: near-complete repair means recovery of the measured gap under the predefined metric, not a universal guarantee.

Figure 3. The reliability panel asks whether lower-energy quantiles are useful. The compute panel asks whether more energy evaluations buy real utility or only better optimization of a bad energy surface. The correct reading is conditional: more compute helps in aligned tails and can harm in miscalibrated tails.

Figure 4. This figure is the simulator honesty check. It separates expert-centered, learned-proposal, state-heuristic, random, and scripted variants. If a reader cares about autonomous learned-policy success, this is the figure that keeps the claim unsupported.

Figure 5. This figure condenses the high- N controlled audit. It is useful when a reviewer asks whether the paper is cherry-picking a single row. The matrix shows which scorer families have negative selected utility at $N = 512$ and which do not.

Figure 6. The left panel checks whether repair recovery clears the threshold across repair models. The right panel checks whether simulator success survives dependency decomposition. Together they prevent the paper from claiming a repair or simulator result without stating its assumptions.

Figure 9. This figure is not decorative. It makes the claim ledger visible. A paper can be strong while keeping unsupported endpoints unsupported. The optional rollout panel makes clear that optional adapters are finite execution artifacts.

Figure 17. This is a maintenance figure. It helps a fresh agent see that the final paper is tied to source, scripts, tests, docs, figures, tables, and generated summaries rather than a manually edited PDF.

A.23 CLAIM TIERS AND UPGRADE POLICY

V4 uses four claim tiers. Tier 1 is exact selected-tail accounting. These claims are strongest because they follow from a finite identity and are validated by Monte Carlo. Tier 1 includes the tie-aware selected-utility law and exact-law validation.

Tier 2 is local EBM evidence. These claims are supported by controlled contact/multimodal tasks, learned NumPy and PyTorch IBC-style EBMs, repair stress tests, reliability diagrams, and compute frontiers. Tier 2 supports the selected-tail mechanism and scoped repair statements, but it does not imply real-robot validation.

Tier 3 is simulator evidence. These claims are tied to Meta-World and optional adapter artifacts. The selected-action benchmark and closed-loop dependency audit are useful but bounded. State-heuristic success is supported; autonomous learned-policy success remains unsupported unless learned proposal variants clear the threshold across tasks without runtime expert or state-heuristic help.

Tier 4 is external robotics validation. This tier would require real hardware or high-fidelity benchmark evidence with fixed splits, candidate-level logs, measured utility, deployment gates fixed before evaluation, and failure rows published. V4 discusses this tier as future work and a benchmark-readiness checklist, not as an achieved result.

The upgrade policy is conservative. A future revision may promote a boundary only if the repository contains artifacts for that endpoint and the claim audit is updated to require them. Prose alone cannot upgrade a claim tier.

A.24 RESOURCE AND REPRODUCIBILITY POLICY

The full experiment stack is CPU-feasible but not cheap. The final audit docs record a long Meta-World refresh and a roughly thousand-second full run. That is acceptable for regeneration, but it is not acceptable as a required step for every paper compile. V4 therefore separates experiment regeneration from submission verification.

Submission verification is the path a fresh session should run first. It executes the V4 cache, rebuilds the PDF, runs the claim audit, checks the tests, verifies hashes, and inspects representative rendered pages. This path does not retrain EBMs or rerun simulators. It keeps memory low by streaming CSV/JSON artifacts and writing compact outputs.

Full regeneration is the path for reviewers who want to rebuild the evidence from code. It runs the controlled tasks, learned EBMs, repair studies, Meta-World ladder, optional adapters, figures, claim audit, and tests. If full regeneration changes artifacts, the V4 cache and manuscript must be rebuilt before the PDF is treated as final. A paper cannot be submission-ready if its PDF cites stale numbers.

A.25 FRESH-SESSION ARCHIVE CONTRACT

A fresh session should use the Desktop source map as the first lookup table. For this paper, the final row must identify `ebm-policy-best-of-n-v4.pdf`, the local folder `C:/Users/wangz/ebm-policy-best-of-n`, and the GitHub repository `Jason-Wang313/ebm-policy-best-of-n`. The folder name is the source of truth. The PDF name and GitHub repository name should point back to that folder without guesswork.

Inside the repository, the final versioned PDF belongs at `paper/final/ebm-policy-best-of-n-v4.pdf`. The anonymous LaTeX build PDF under `paper/iclr2026/build/main.pdf` is a local build product, not the versioned artifact. The Desktop PDF must match the repository final PDF byte-for-byte.

The archive contract also requires that old Desktop v2 PDFs be absent or superseded. If a future agent sees both v2 and V4 on the visible Desktop, it may choose the wrong file. V4 therefore treats stale Desktop versions as an audit failure.

A.26 HOW TO REJECT OR ACCEPT CORRECTLY

A correct rejection should say that the controlled and simulator evidence is not enough for the desired venue or claim tier. It should not say the paper claims real-robot validation, autonomous learned-policy success, or broad manipulation success, because V4 explicitly blocks those claims. It should not say high N always hurts, because the paper reports aligned and oracle cases where high N helps or saturates.

A correct weak-accept argument should say that EBM inference-time scaling has a deployment-tail failure mode that is under-discussed, and that the paper provides a reproducible audit stack with exact accounting, controlled failures, learned EBM artifacts, scoped repairs, and simulator dependency boundaries. The accept argument should still mention the missing real-robot validation and autonomous learned-policy limitation.

A correct revision request should ask for one of three concrete upgrades: larger learned EBM models with candidate-level logs, a learned proposal that clears the dependency threshold across tasks, or real/higher-fidelity robot validation with fixed gates and candidate-pool oracle metrics. A request for "more experiments" without specifying the claim tier would be less useful.

A.27 EXTERNAL VALIDATION PROTOCOL

An external validation should begin by freezing tasks, observations, candidate generators, energy scorers, and gates. The selected-tail question becomes ambiguous if the proposal generator changes after seeing tail failures. The candidate pool should be saved before selection so the finite law, oracle gap, support diagnostics, and selected-tail reliability can be recomputed later.

For each task, the validation should report N -sweeps rather than a single best N . It should include selected energy, selected utility, low-energy tail utility, invalid or unsafe action rate, support distance, success, oracle gap, repair recovery, and gate decision. It should also report failures and blocked high- N cases, because a safe gate is supposed to block some cases.

For learned EBMs, the validation should separate training objective, proposal source, action refinement, and selection. If a CEM or Langevin refinement procedure is used, its compute budget and latency proxy should be logged beside N . Otherwise a result could improve because the proposal improved while selected-tail calibration worsened, or vice versa.

For robot or high-fidelity simulator tasks, the validation should include closed-loop variants without scripted continuation and without runtime expert centering. Expert demonstrations can be used for training, but deployment claims should identify whether candidate proposals still depend on expert actions, nearest-demo retrieval, behavior cloning, state heuristics, or local Gaussian sampling.

A.28 WHAT V4 INTENTIONALLY DOES NOT PROVE

V4 does not prove that EBMs are bad robot policies. The aligned PyTorch EBM and oracle/good scorer rows are included to block that reading. The paper proves that a selected-tail audit is necessary when the policy deploys by optimizing energy over candidates.

V4 does not prove that more candidates are bad. More candidates are useful when the low-energy tail has high real utility. The failure is conditional on tail miscalibration.

V4 does not prove that support penalties or calibration always fix selected tails. The repair stack is strong under the measured local stress tests. It is not a universal method, and it is not a safety certificate under distribution shift.

V4 does not prove broad manipulation success. The simulator ladder and optional adapters are useful artifacts, but their boundaries are explicit. Real-robot validation remains absent.

V4 does not claim the finite law as a standalone novelty. The law is an audit instrument. The EBM-specific contribution is the low-energy-tail framing, the diagnostic stack, and the dependency-aware evidence package.

A.29 FAILURE-CASE WALK-THROUGHS

The strongest controlled failure can be read as a five-step story. First, the candidate generator proposes a finite action pool. Second, the raw miscalibrated energy gives very low energy to a subset of actions that are invalid or far from support. Third, increasing N increases the probability that at least one of those low-energy bad actions appears in the sampled set. Fourth, the deployment rule selects that low-energy action. Fifth, real utility falls even though the energy objective has improved. The selected energy -2.988 and selected utility -1.538 in the strongest row are a compact instance of this story.

The repair walk-through is different. The repair does not change the finite candidate pool. It changes the score used to select from that pool by adding calibration, support information, and a conservative high- N rule. If the repair works, it moves selection away from unsupported low-energy shortcuts toward actions whose measured utility is closer to the in-pool oracle. This is why repair recovery is normalized by the raw-to-oracle gap. Without that normalization, a repair on an easy pool and a repair on a hard pool would be hard to compare.

The dependency failure is subtler. A selected-action benchmark can show high reward because the candidate proposals are near expert actions or because scripted continuation rescues a weak first action. In that case the energy scorer may still be useful, but the paper cannot call the result autonomous. The closed-loop audit makes this visible by comparing proposal families. The state-heuristic success result is therefore reported as state-heuristic low-dependency simulator evidence, not as learned-policy autonomy.

The optional adapter walk-through is the narrowest. A finite adapter rollout may report reward and even nonzero scripted success. That proves the adapter ran and the logging path works. It does not prove that a learned EBM policy solved a suite. V4 includes optional adapters to prevent future confusion: rows with zero success or unavailable success remain visible rather than being silently omitted.

A.30 ROBUSTNESS QUESTIONS

Does the conclusion depend on one seed? The controlled artifacts use 5 seeds and 9 high- N summary rows. The paper should not claim a large statistical population, but it is also not a single anecdotal rollout. The seed-level tables remain in the repository for reviewers who want to check heterogeneity.

Does the conclusion depend on one scorer? No. The controlled audit includes oracle, good, raw miscalibrated, shortcut, smoothness-blind, out-of-support-low-energy, support-blind, noisy, and pilot-label-noise scorers. The point is comparative: aligned tails behave differently from miscalibrated tails.

Does the conclusion depend on one model implementation? No. The repository contains analytic/scored controlled artifacts, a NumPy IBC-style learned EBM, and a PyTorch MLP EBM. The PyTorch model is important because it has a strong aligned-tail result rather than simply reproducing the failure.

Does the conclusion depend on one benchmark? The strongest mechanistic evidence is controlled, not benchmark-specific. The simulator ladder and optional adapters are secondary evidence. They broaden the artifact surface while preserving boundaries.

Does the conclusion depend on one repair? No. The repair hierarchy includes support-only, calibrated, value-shaped, conservative-gated, and oracle or near-oracle variants. The main repair stack is promoted only because it clears the metric-bound thresholds while weaker deployable ablations and stronger upper bounds remain visible.

A.31 TABLE-LEVEL SANITY ASSERTIONS

The controlled summary table should satisfy three sanity checks. The oracle or good rows should not exhibit the same high- N collapse as raw miscalibrated rows. The raw miscalibrated rows should show lower selected energy at high N , because the optimizer is in fact finding lower-energy actions. The selected utility should be evaluated by the external utility, not by the energy itself.

The reliability table should contain extreme-tail rows. If it only reported average buckets, it would miss the deployment object. V4 records 80 extreme-tail rows. A reviewer can therefore inspect whether the very low-energy tail has high utility, low invalid rate, and reasonable support distance.

The repair table should report failures and utility-degrading rows. A repair paper that hides failures is not audit-ready. V4 reports 1 diagnostic repair failure and 0 utility-degrading rows for the main repair stack. The failure count is what prevents the paper from drifting into universal repair language.

The closed-loop table should include variants that can fail. Random uniform and local Gaussian variants are necessary because they show that success is not automatic. Learned-demo and behavior-cloned variants are necessary because they test whether a learned proposal can replace state heuristics or expert-centered candidates. State-heuristic rows are necessary because they show a lower-dependency simulator path that is still not a learned-policy claim.

The optional adapter table should include zero-success rows and unavailable success rows. Otherwise optional benchmarks become marketing evidence rather than scientific evidence. V4 treats them as finite guarded rollouts.

A.32 AUTHOR-RESPONSE TEMPLATES

If asked for more robotics. We agree that real-robot validation would strengthen and broaden the paper. The current claim is intentionally scoped to selected-tail accounting, controlled and learned EBM artifacts, and guarded simulator/dependency audits. The paper includes a benchmark-readiness checklist specifying the candidate logs, utility measurements, proposal-dependency controls, and gates needed to upgrade the claim tier.

If asked why exact laws matter. The law lets us separate two questions. One question is whether an EBM scores actions well. The other is what selected utility follows from a fixed score, proposal, and utility pool under high- N inference. The law answers the second question exactly, including ties, so later curves are not merely noisy Monte Carlo anecdotes.

If asked whether the repair is heuristic. Yes, the repair is a practical diagnostic stack, not a universal theorem. Its claim is metric-bound: on supported local stress tests, the combined repair recovers most of the raw-to-oracle gap while reporting failures and avoiding utility-degrading rows. We do not claim the same repair will work under unobserved hidden modes or distribution shift without new evidence.

If asked whether this duplicates other tail papers. The shared finite law is a measurement tool. The paper identity is EBM action inference: conditional action energies, low-energy action tails,

support penalties, IBC-style learned EBMs, compute/refinement frontiers, and closed-loop proposal-dependency audits. These are not WAM rollouts, JEPA latent predictions, diffusion trajectories, or LLM answer reranking.

If asked whether the simulator result is autonomous. No. The selected-action benchmark uses scripted continuation, and the closed-loop audit shows that learned-proposal variants do not clear the learned-policy threshold. The supported simulator claim is dependency-aware: state-heuristic low-dependency variants clear the threshold, while autonomous learned-policy success remains unsupported.

A.33 FINAL ADVERSARIAL CHECKLIST

Before accepting the paper’s own claims, a reviewer should be able to answer the following questions from artifacts:

1. Which rows show energy improving while utility collapses?
2. Which rows show aligned tails where high N helps or saturates?
3. What is the exact-law validation error?
4. What is the denominator for repair recovery?
5. Which repair rows are deployable and which are upper bounds?
6. Which simulator rows use scripted continuation?
7. Which closed-loop rows depend on expert-centered candidates?
8. Which learned-proposal rows fail the autonomous learned-policy threshold?
9. Which optional adapters have zero or unavailable success?
10. Which claims are supported, and which endpoints are non-claim boundaries?
11. Can the paper be rebuilt without rerunning the heavy simulator stack?
12. Do the repo PDF and Desktop PDF have the same hash?

If any answer is missing, the V4 artifact is not submission-ready. If all answers are present, the remaining debate is about claim tier and venue fit, not about whether the paper hid its evidence.

A.34 FRESH-AGENT INVARIANTS

The final artifact is designed for a future session that has no memory of this conversation. That session should be able to verify five invariants. First, the Desktop source map names the final PDF, the local folder, and the GitHub repository. Second, the repository contains the final V4 PDF under `paper/final`. Third, the Desktop PDF and repository PDF have identical hashes. Fourth, the V4 audit script can regenerate the evidence cache and confirm the page count. Fifth, the remote GitHub branch points to the commit that contains the final PDF and scripts.

The second invariant is especially important because the LaTeX build directory is not the archive. A local build product can be deleted and recreated. The final V4 PDF is the committed artifact a reviewer or future agent should compare against the Desktop copy. This avoids the common error of submitting an old build PDF while the repository source has moved on.

The third invariant prevents invisible drift. If the Desktop PDF has a different hash from the repository PDF, one of them is stale. The correct response is to rebuild, copy, and rerun the audit, not to guess which file is newer from modification times.

The fourth invariant keeps the evidence tied to the manuscript. If a future experiment regeneration changes selected utility, repair recovery, or dependency rows, the V4 cache will change. The macros and figures must then be rebuilt before the paper is final again. This is why the V4 manuscript cites generated macros instead of manually typing every headline number.

The fifth invariant makes GitHub part of submission readiness. A local PDF on Desktop is not enough. The source folder must be pushed so a fresh session can recover the scripts, artifacts, and final PDF from the repository. The final audit should therefore include both local checks and remote SHA verification.

A.35 REVIEWER SCORECARD

A reviewer who values theoretical cleanliness should give credit for the finite tie-aware selected-utility law, but should not treat the law as the paper’s only contribution. The law earns its place because it makes the selected-tail curves exact for fixed pools.

A reviewer who values empirical rigor should check whether the paper reports negative controls, positive controls, learned artifacts, repair ablations, dependency audits, and optional adapter boundaries. V4 exposes all of these. The remaining weakness is scale and external validation, not absence of local artifact rigor.

A reviewer who values robotics relevance should focus on the dependency audit. The paper is relevant because EBM inference is action optimization, but it is not a real-robot paper. The simulator evidence should be scored as scoped simulator evidence with explicit proposal boundaries.

A reviewer who worries about duplicate submissions should compare the scientific object. This paper’s object is low-energy action selection under $E_\theta(o, a)$. Its diagnostics include support distance, invalid actions, IBC-style EBMs, compute/refinement budgets, and proposal-dependency audits. Those are the identity features that separate it from other tail-selection papers.

A reviewer who worries about overclaiming should inspect the non-claim boundaries. V4 does not hide unsupported endpoints. It lists them and makes the claim audit fail if promoted prose claims them.

The final scorecard is therefore mixed but defensible: strong mechanism, strong artifact discipline, clear repair metrics, honest simulator boundaries, missing real robot, and no autonomous learned-policy claim. That is the paper the V4 artifact actually supports.

A.36 REPRODUCIBILITY COMMANDS

The anonymous paper PDF can be rebuilt from the repository root with:

```
python scripts/build_v4_paper.py
```

The final claim audit can be run with `python scripts/run_v4_claim_audit.py`. The test suite can be run with `python -m pytest -q`. Long simulator refreshes are intentionally not required for rebuilding the paper because the paper consumes the existing CSV/JSON/PNG artifacts.